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King Abdullah University of
Science and Technology

Tile-Low Rank Compressed Multi-Dimensional Convolution and Its Application to Seismic Redatuming Problems

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Authors	Ravasi, Matteo; Hong, Y.; Ltaief, Hatem; Keyes, David E.
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Introduction

A broad class of seismic data processing and imaging algorithms require the repeated evaluation of the so-called Multi-Dimensional Convolution (MDC) operator and its adjoint (Ravasi and Vasconcelos, 2021). Examples of such kind are Estimation of Primaries by Sparse Inversion, Multi-Dimensional Deconvolution, and Marchenko-based redatuming and demultiple. A common challenge in the application of these techniques to 3D seismic data is represented by the poor scalability of the MDC operator due to its inherent memory-bound behavior. The advent of massively parallel architectures represents an opportunity for researchers in the geophysical community. Nevertheless, the underlying algorithms are required to embrace a programming paradigm shift to fully benefit from hardware advances, the main reason being the continuing decrease in ratio between memory bandwidth and processing power that renders data motion as the main performance bottleneck (Unat et al., 2017).

In this work, we revisit the implementation of the MDC operator in the context of 3D seismic data applications and identify two key strategies that can reduce its overall memory footprint whilst minimally impacting on the quality of the modelling operator and associated inverse problem, namely Tile Low-Rank (TLR) compression and geographically aware re-arrangement of the kernel matrices. The proposed strategies are validated on a synthetic example of 3D Marchenko redatuming.

Acceleration strategies for MDC operator

The MDC operator can be expressed by the following integral relation:

$$y(t, \mathbf{x}_S, \mathbf{x}_F) = \mathcal{F}^{-1} \left(\int_{\delta\mathbb{D}} R(f, \mathbf{x}_R, \mathbf{x}_S) \mathcal{F}(x(t, \mathbf{x}_R, \mathbf{x}_F)) d\mathbf{x}_R \right), \quad (1)$$

where $R(f, \mathbf{x}_B, \mathbf{x}_R)$ represents the seismic reflection response in the frequency-space domain (also referred here as the kernel of the MDC operator), whilst $x(t, \mathbf{x}_R, \mathbf{x}_F)$ and $y(t, \mathbf{x}_S, \mathbf{x}_F)$ are the input and output seismic wavefields. Here $\mathcal{F}$ and $\mathcal{F}^{-1}$ are the forward and inverse Fourier transforms, t is time, f is the angular frequency, $\mathbf{x}_F$, $\mathbf{x}_R$ and $\mathbf{x}_S$ identify the spatial locations of the subsurface focus point, surface receivers, and surface sources, respectively. Once discretised, the integral in equation 1 must be implemented as a series of matrix-vector multiplications (MVMs), also referred to as a batched MVM. As the kernel of the MDC operator $R(f, \mathbf{x}_B, \mathbf{x}_R)$ is obtained by applying a Fourier transform over the time axis of the recorded seismic reflection response $R(t, \mathbf{x}_B, \mathbf{x}_R)$, this inherently results in a stack of N complex-valued dense matrices $\{\mathbf{R}_{f_1}, \mathbf{R}_{f_2}, \dots, \mathbf{R}_{f_N}\}$, where $\mathbf{R}_{f_i} \in \mathbb{C}^{N_S \times N_R}$.

Tile-low rank compression

Low-rank matrix approximation is a class of algebraic compression methods that aim to exploit data sparsity in the matrix to be compressed. Contrary to other compression techniques, such as the ZFP algorithm adopted in the context of seismic processing by Brackenhoff et al. (2021), low-rank approximations allow matrix-vector operations to be directly applied on the compressed matrix without the need for decompression. By doing so, the algorithmic complexity and memory footprint of matrix vector multiplication (MVM) can be greatly reduced (Grasedyck, 2013). The algorithmic design of Tile Low-Rank Matrix-Vector Multiplication (TLR-MVM) is briefly presented here – see Ltaief et al. (2021) and Hong et al. (2021) for more details. As shown in Figure 1a, each dense matrix $\mathbf{R}_{f_i}$ is split into tiles of size $N_b \times N_b$ and the low-rank nature of such tiles is revealed using an algebraic method of choice (e.g., rank-revealing QR, randomized-SVD). More specifically, every tile $\mathbf{R}_{f_i}^{(j,k)}$ is decomposed into a pair of $\mathbf{U}^{(j,k)} \in \mathbb{C}^{N_b \times K^{(j,k)}}$ and $\mathbf{V}^{(j,k)} \in \mathbb{C}^{K^{(j,k)} \times N_b}$ bases such that $\|\mathbf{R}_{f_i}^{(j,k)} - \mathbf{U}^{(j,k)} \mathbf{V}^{(j,k)}\|_F \leq \varepsilon \|\mathbf{R}_{f_i}\|_F$. Note that the ranks $K^{(j,k)}$ of the different tiles within the same matrix can be different. Finally, the actual TLR-MVM operation is applied for each frequency in the following three consecutive phases: (1) the various $\mathbf{V}^{(j,k)}$ bases are stacked vertically and multiplied to an input vector $\mathbf{x} \in \mathbb{C}^{N_R}$, (2) the result from phase 1

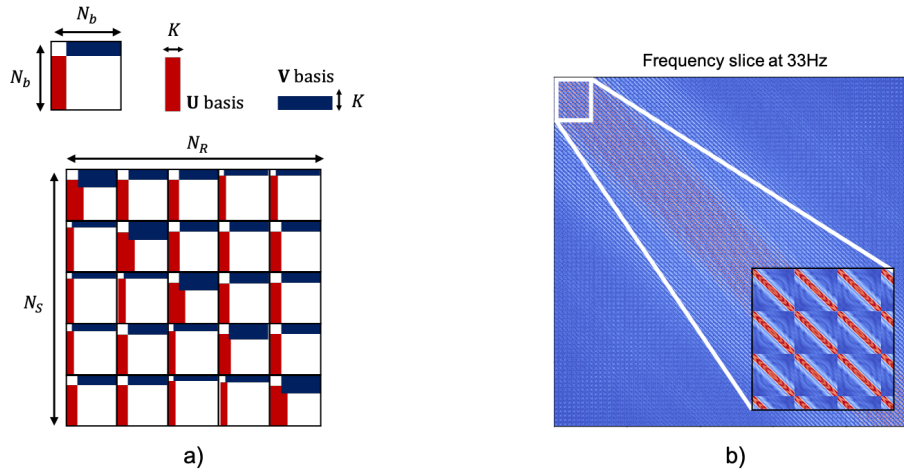


Figure 1 a) Schematic representation of the TLR compression algorithm. b) Frequency kernel at $f = 33\text{Hz}$ and close-up showcasing its block structure inherited by the arrangement of sources and receivers in the original acquisition geometry.

is reshuffled, and (3) horizontally stacked $\mathbf{U}^{(j,k)}$ bases are multiplied to the intermediate vector to obtain the output vector $\mathbf{y} \in \mathbb{C}^{N_s}$.

Geographically aware kernel sorting

A seismic reflection response, used here as the kernel of the MDC operator, can be conceptually visualized as a three-dimensional tensor of size $N_f \times N_s \times N_r$, where frequency is arranged along the slowest axis. When performing MDC with dense frequency matrices, the arrangement of sources (along the rows of each matrix) and receivers (along the columns of each matrix) can be totally arbitrary as long as consistent with that of the input vector $\mathbf{x}$. Such an arrangement is however much more important within the context of TLR compression as it may lead to better or worse block compression capabilities. A variety of re-arrangement strategies that aim at reducing the overall distance between sources (and receivers) within the same tile have been investigated. Two options that can drastically reduce the ranks of the different tiles have been identified. The first, referred to as L1 sorting, re-arranges sources in groups of N_b nearest sources based on the L1 norm between the most bottom-left source and all the other available sources (i.e., sources that are not yet being assigned to a group) (Figure 2b). Alternatively, the Hilbert sort algorithm, also known as Hilbert space-filling curve, is used (Figure 2c).

Numerical results

The proposed strategies are tested on the 3D synthetic dataset presented in Ravasi and Vasconcelos (2021). This dataset was generated in a constant-velocity, varying-density model using a dense grid of 9801 co-located sources and receivers for a maximum time of $T = 1.5\text{s}$. The overall size of the dataset, stored in single-precision floating point format, is 230 GigaBytes (GB). As commonly done in seismic processing, sources and receivers are arranged along parallel *cable* lines as shown in Figure 2a. In a pre-processing phase, the data is converted to the frequency domain and the first 150 frequency components ($0 < f < 50\text{Hz}$) are selected to form the complex-valued kernel of the MDC operator whose size is 108 GB. A close-up of the first few hundred rows and columns of a frequency matrix reveals a block structure (Figure 1b), a consequence of the source-receiver arrangement; more importantly, each block is diagonally dominant allowing a successful low-rank approximation of the seismic data. As a result, the collection of $\mathbf{U}$ and $\mathbf{V}$ bases for the entire frequency spectrum amounts to a size of 29 GB (approximately four times smaller than the original stack of dense matrices). Whilst this result already proves the benefit of TLR compression, the various frequency matrices are now re-arranged following the *L1* and *Hilbert* arrangements. Figures 2d-f display heatmaps of the ranks of tiles for the original and re-arranged kernel matrices for frequency $f = 33\text{Hz}$. Hilbert (and to a certain extent L1) sorting

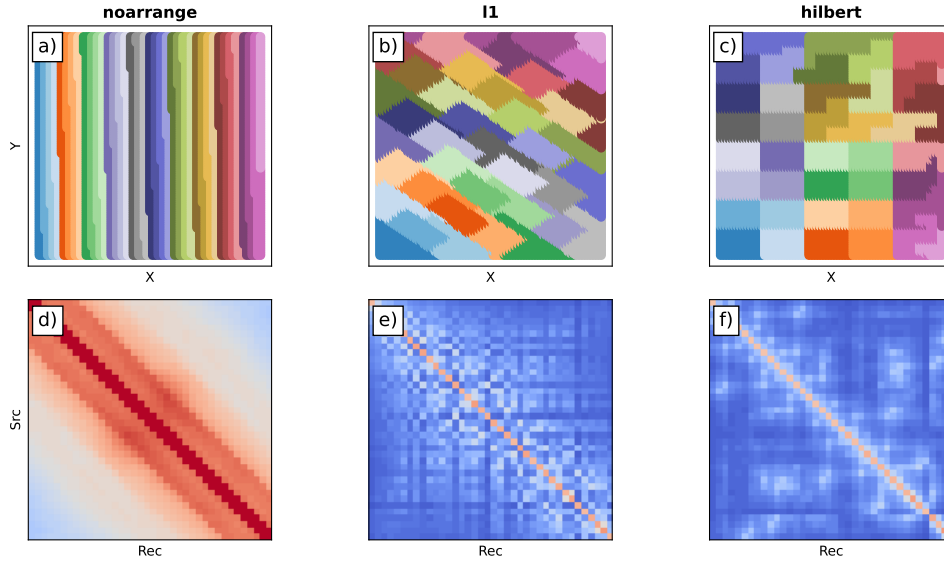


Figure 2 Geographically aware kernel arrangement. a-c) Source (or receiver) geometries color-coded on groups of size $N_b = 256$ for the cases of no-arrangement, arrangement based on L1 norm criterion, and arrangement based on Hilbert sorting, respectively. d-f) Corresponding tile ranks for a frequency kernel at $f = 33\text{Hz}$ (red: high rank, blue: low rank).

have been able to condense most of the independent information in the matrix around the main diagonal, whilst off-diagonal tiles have much lower ranks compared to those of the original arrangement.

The different flavours of TLR-compressed MDC kernels are finally used as input to the Marchenko redatuming algorithm and the resulting Green's functions are compared to that obtained from the dense kernel in Figure 3a. The error arising from TLR approximation of the MDC kernel is minimal in all cases (Figures 3b-d). Finally, the performance of dense MVM and TLR-MVM using both the original and Hilbert arrangements for the same frequency matrix discussed above are compared across a variety of hardware (Figure 4). The experiments are conducted on x86_64 platforms (Intel Ice Lake, AMD EPYC Rome), ARM (Fujitsu A64FX) and hardware accelerators (NVIDIA A100 80GB, NEC Aurora Vector Engines). We report both memory bandwidth and time to solution, since TLR-MVM is a memory-bound algorithm. As shown in Figure 4b, we obtain a speed-up of about 36.1, 203.5, 26.4, 4.8, and 7.3 (from left to right) for TLR-MVM with Hilbert sorting compared to their equivalent dense MVM, respectively. Bandwidth results in Figure 4a show that TLR-MVM can achieve better bandwidth due to smaller data size and contiguous memory layout. On AMD EPYC Rome, we achieve bandwidth that is higher than DRAM sustained bandwidth thanks to large last level cache (512MB) of the hardware.

Conclusions

We have proposed a set of strategies to alleviate the heavy memory requirements and computational cost associated with the application of multi-dimensional convolution operators in the context of seismic processing and imaging. By re-arranging the rows and columns of frequency-domain seismic data and exploiting their low-rank nature, we have been able to reduce the memory footprint by a factor of 16 leading to speed-ups in the order of 4.8 to 203.5 on various hardware. Showcased with an example of 3D Marchenko redatuming, our findings have broader implications for the seismic community in that they can be also applied to any of the above mentioned algorithms that rely on the MDC operator.

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TLR-MVM benchmarks can be accessed at <http://github.com/ecrc/tlrmvm>.

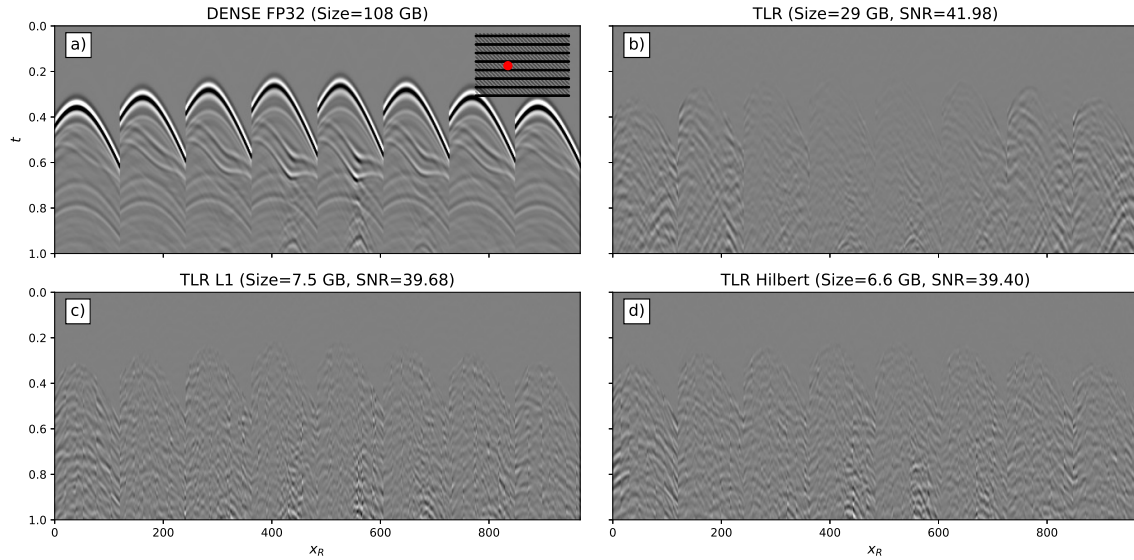


Figure 3 a) Green's function estimate using the dense kernel. Error ($\times 10$ clip) for TLR compressed estimates using b) original matrix arrangement, c) L1-based re-arrangement, d) Hilbert-based re-arrangement. The sizes of each kernel and reconstruction SNRs are shown in the title of each panel.

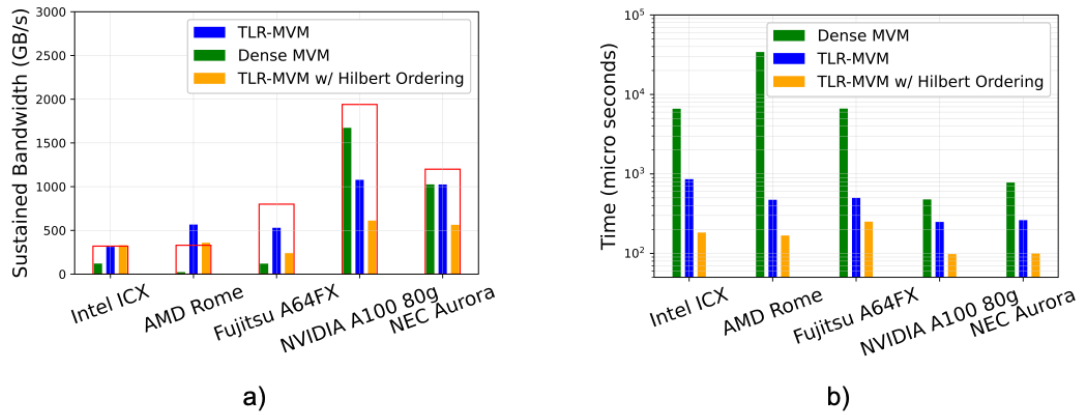


Figure 4 Performance benchmarks for dense MVM and TLR-MVM on various hardware. a) sustained bandwidth, and b) time to solution.

References

- [1] Brackenhoff, J., Thorbecke, J., Meles, G., Koehne, V., Barrera, D., and Wapenaar, K. [2021] 3D Marchenko applications: Implementation and examples. Geophysical Prospecting.
- [2] Grasedyck, L., Kressner, D., and Tobler, C. [2013] A Literature Survey Of Low-Rank Tensor Approximation Techniques GAMM-Mitteilungen, 36, 53–78.
- [3] Hong, Y., Ltaief, H., Ravasi, M., Gatieneau, L., and Keyes D.E. [2021] Accelerating Seismic Redatuming Using Tile Low-Rank Approximations on NEC SX-Aurora TSUBASA. Supercomputing Frontiers and Innovations, 8.
- [4] Ltaief, H., Cranney, J., Gratadour, D., Hong, Y., Ferreira, F., and Keyes, D. [2021] Meeting The Real-Time Challenges of Ground-Based Telescopes Using Low-Rank Matrix Computations, Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis.
- [5] Ravasi, M., and Vasconcelos, I. [2021] An open-source framework for the implementation of large-scale integral operators with flexible, modern hpc solutions - Enabling 3D Marchenko imaging by least-squares inversion. Geophysics, 86, WC177–WC194.
- [6] Unat, D., et al. [2017] Trends in Data Locality Abstractions for HPC Systems. IEEE Transactions on Parallel and Distributed Systems, 28, 3007–3020.